Fairness-Aware Active Learning for Decoupled Model

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***Abstract*—Active learning (AL) aims to actively label training data for supervised learners to efficiently improve accuracy of the trained models. However, traditional AL techniques ignore model fairness, which is an important social norm in modern machine learning applications. In this paper, we propose the first fairness-aware active learning technique for decoupled model called D-FA^{2}L. It labels data that receive disagreed predictions from the decoupled models, and we hypothesize it can simultaneously improve model fairness and model accuracy. We present theoretical justification of D-FA^{2}L, proving it can reduce model unfairness under proper conditions. In experiments, we show D-FA^{2}L can reduce model unfairness effectively, and the reduction is more efficient than both classic and state-of-the-art AL techniques on three data sets. In the meantime, D-FA^{2}L is able to maintain similar improvement rate on model accuracy.**

***Index Terms*—Active Learning, Fairness, Decoupled Model**

I. Introduction

Active learning (AL) is a popular research topic that offers sample-efficient ways to train models [1]. A typical AL scenario assumes one needs to label training data before feeding them into a supervised learner, but labeling comes with cost so one aims to reduce the total cost by actively labeling as few data as possible while maximizing accuracy of the trained model. Although many active labeling strategies have been successfully applied in different domains e.g. [2], [3] and shown to improve model accuracy more efficiently than random labeling, few of them consider the impact on model fairness.

Fairness in machine-learned models has become an important aspect to consider in modern machine learning applications. Take AI-aided recruitment as an example, where a model is used to automatically score human candidates. Ideally, a model without gender bias should not systematically lower score female candidates, and a model with no racial bias should not systematically lower score black candidates. In reality, however, standardly learned models often exhibit bias. For example, the widely used COMPAS recidivism prediction model is shown to have racial bias [4] and Amazon's hiring model is shown to have gender bias [5]. These discoveries have motivated intensive research that aims to incorporate fairness in machine learning techniques; see e.g. [6]–[11] and reference therein.

How to incorporate fairness in active learning is an emerging research topic. To our knowledge, a state-of-the-art labeling strategy is FAL [12], which labels data that receive uncertain predictions and are expected to reduce model unfairness if added to the training set. This strategy is designed for single model, and its effectiveness on the decoupled model studied in this paper remains unknown. Decoupled modeling [13] is a promising technique which develops different models for different groups, respectively, and admits an efficient fairness-accuracy trade-off. Current decoupled model learning assumes training data are randomly labeled, however, and does not consider sample efficiency as active learning.

In this paper, we propose the first fairness-aware active learning technique for decoupled model called D-FA^{2}L, which labels data disagreed by the decoupled models. Our hypothesis is the technique can improve model accuracy and model fairness simultaneously. We verify this hypothesis from two directions. On the theory side, we prove D-FA^{2}L can reduce model fairness under proper conditions. In experiments, we show D-FA^{2}L reduces model unfairness more effectively and efficiently than both classic and state-of-the-art AL techniques, while being able to maintain a similar accuracy on three data sets.

To summarize, the contributions of this paper include

- Propose the first fairness-aware active learning technique for decoupled model (D-FA^{2}L) and show it can reduce model unfairness effectively
- Show D-FA^{2}L can reduce model unfairness more efficiently than both classic and state-of-the-art AL techniques
- Theoretical justify on the efficacy of D-FA^{2}L

The rest of this paper is organized as follows. We discuss the background in Section 2, and present the D-FA^{2}L algorithm in Section 3. We then present theoretical justification of D-FA^{2}L in Section 4 and experimental results in Section 5. In Section 6, we conclude the present study.

II. Background

A. Fairness in Machine Learning

Fairness has become an important societal aspect to consider in modern machine learning applications, and how to incorporate it in machine-learned model has been extensively studied in the research community e.g., [14]–[16]. The proposed techniques can be grouped into three categories: (i) data pre-processing, (ii) model regularization and (iii) model post-processing. In (i), one modifies training data before feeding them into standard learning algorithm [17], [18]. In (ii), one adds a properly designed fairness constraint to the learning process requiring the model to minimize both prediction loss and unfairness e.g. [19], [20]. In (iii), one post-processes a standardly learned model to achieve fairness e.g. [21], [22].

Despite the success of many fairness-aware learning techniques, most of them focus on the passive setting where training

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data are randomly labeled. This is not very sample-efficient as data labeling is often an expensive process associated with certain cost. To reduce the total cost, it is more desirable to perform active learning that strategically labels training data. In this paper, we study fairness-aware active learning.

B. Active Learning

Active learning (AL) is a long-studied research topic in machine learning [1]. In a typical AL scenario, it is assumed that labels of training data for supervised learning are expensive and time-consuming to collect e.g., one may need to hire human experts to manually evaluate massive data and label them. As a result, labeling each instance comes with a cost. To minimize the total cost, AL aims to label as few training data as possible while maximizing accuracy of the trained model.

Many active labeling strategies are proposed in the literature. A classic example is uncertainty-based, which labels data that receive the most uncertain predictions from the current model. Another example is disagreement-based, which labels data that receive disagreed predictions from a committee of models. The basic hypothesis of AL is that actively labeled training data can improve model accuracy more efficiently than randomly labeled training data, and thus active learning is more sample-efficient than passive learning (i.e. learning with randomly labeled data).

Although AL techniques have been successfully applied in many domains e.g. [2], [3], few of them consider the fairness aspect in the trained models. As pointed out in [12], classic AL technique can improve a model's accuracy at the cost of deteriorating its fairness. Our hypothesis is such trade-off can be significantly reduced with properly designed fairness-aware active labeling strategy, as the one proposed in this paper.

C. Fairness in Active Learning

Fairness-aware active learning is an emerging research topic which aims to label training data for improving model accuracy while maintaining or even improving model fairness. In [12], a novel algorithm called fair active learning (FAL) is proposed. It actively labels data that are both uncertain to the model and can reduce its unfairness in expectation, and then feeds these data into a standard supervised learner to train models. A different setting is studied in [23], where a meta-learning algorithm is proposed that uses stochastic gradient descent to iteratively update a model using actively selected training data. In this paper, we focus on the setting in [12]. Moreover, all prior studies focus on active learning for single model, while we focus on active learning for decoupled model. Our experimental results suggest the prior fairness-aware active labeling strategy may not be as effective on the decoupled model as it is on the single model.

III. Proposed Algorithm

In this section, we propose a disagreement-based fairness-aware active learning strategy (D-FA^{2}L) for decoupled model. We begin with the problem setting.

Let (x, s, y) be an instance, where x is the vector of non-protected features, s is a binary protected feature, and y is a binary label. A decoupled model is a pair of models (h_0, h_1) such that h_i is applied on instances with $s = i$. A standard decoupled supervised learner $\mathcal{A}_d$ trains model h_i from instances with $s = i$ using any standard supervised learning technique.

Algorithm 1 Disagreement-based FA^{2}L Algorithm (D-FA^{2}L)

Input: an initial labeled training set L, a pool of unlabeled data set U, a hypothesis class H, a standard decoupled supervised learner $\mathcal{A}_d$, hyper-parameters α, k.
Output: a decoupled model (h_0, h_1).
1: Train $(h_0, h_1) \in H \times H$ from L using $\mathcal{A}_d$.
while Stopping criterion is not met **do**
2: Independently and uniformly select k instances $u \in U$ satisfying $|h_0(u) - h_1(u)| > \alpha$
3: Label the selected instances, add them to L, and remove them from U.
4: Retrain (h_0, h_1) from L using $\mathcal{A}_d$.
end while

We focus on the classification problem, but assume model h_i will output the posterior probability as $h_i(x) = \Pr\{y = 1 \mid x\}$ instead of directly outputting the binary class – we can obtain the latter by thresholding model output as $\mathbb{1}_{h(x)>0.5}$, where $\mathbb{1}$ is an indicator function.

The basic idea of our proposed strategy is to query label for instance u that receives significantly different predictions from the decoupled models h_0 and h_1, i.e., $|h_1(u) \neq h_0(u)| > \alpha$ for some preset threshold α. This is motivated by the classic disagreement-based active labeling strategy [24], and can be viewed as its extension from single model to decoupled model. Detailed connection and difference are discussed later.

The proposed D-FA^{2}L process is presented in Algorithm 1. In practice, we can stop labeling when a preset number of labels are queried or a desired model performance is achieved.

IV. Theoretical Analysis

In this section, we theoretically analyze D-FA^{2}L. Our main insight is that model unfairness is bounded by a notion of α-distance, which can be efficiently reduced by D-FA^{2}L under proper conditions. In the rest of this section, we first introduce a set of definitions, then present the main theoretical results, and finally discuss the impact on model accuracy.

A. Notations and Definitions

Recall (x, s, y) denotes an arbitrary instance. Let (X, S, Y) be the random variable from which (x, s, y) is sampled. Let μ_* be the joint sampling probability distribution on (X, S) that induces $\mu(x) = \mathbb{P}\{X = x\}$ and $\mu_s(x) = \mathbb{P}\{X = x \mid S = s\}$.

Let H be a hypothesis class from which the decoupled models are learned. Recall each model $h \in H$ is defined as

$$h(x) = \mathbb{P}\{Y = 1 \mid X = x,\ h\}, \tag{1}$$

where randomness comes from the prediction uncertainty of h on a fixed x. The predicted label of x is $\mathbb{1}_{h(x)>0.5}$.

Recall (h_0, h_1) denotes a decoupled model. We evaluate its fairness based on a popular notion called disparate impact [25]. Specifically, for any (h_0, h_1), define

$$\mathtt{DI}(h_0, h_1) = \frac{\mathbb{P}\{h_0(x) > 0.5 \mid S = 0\}}{\mathbb{P}\{h_1(x) > 0.5 \mid S = 1\}}. \tag{2}$$

If $\mathtt{DI}(h_0, h_1)$ is closer to 1, then (h_0, h_1) is more fair.

We will analyze how D-FA^{2}L helps to learn a model with small $\mathtt{DI}(h_0, h_1)$. Define the α-distance of (h_0, h_1) as

$$d_\alpha(h_0, h_1) = \mathbb{P}\{|h_0(x) - h_1(x)| > \alpha\}, \tag{3}$$

where randomness comes from the uncertainty of x w.r.t. μ. An estimate of the distance on a sample J is

$$d_\alpha(h_0, h_1; J) = \frac{1}{|J|} \sum_{x \in J} \mathbb{1}\{|h_0(x) - h_1(x)| > \alpha\}. \tag{4}$$

We will assume the decoupled model returned by a standard learner $\mathcal{A}_d$ has the following form of large margin.

Definition 1. *We say (h_0, h_1) has an α-margin of γ if*

$$\mathbb{P}\{|h_0(x) - h_1(x)| > \alpha \mid \mathbb{1}_{h_0(x)>0.5} \neq \mathbb{1}_{h_1(x)>0.5}\} \geq \gamma. \tag{5}$$

Intuitively, a large α-margin means the decoupled model makes confident prediction on most instances. This is a reasonable assumption e.g., it is not hard to show a standard large margin learner that achieves hard margin α on h_0 and h_1 promises $\gamma = 1$, if we treat $|h_i(x) - 0.5|$ as the margin.

Our analysis will also involve the distance between the data distributions of the two groups, defined as follows.

Definition 2. *The relative total variation distance between distributions μ_0 and μ_1 on any c-weighted data set is*

$$\lambda_c := \max_{\mathcal{Q} \in \Omega_c} \frac{|\mu_0(\mathcal{Q}) - \mu_1(\mathcal{Q})|}{|\mu_0(\mathcal{Q}) + \mu_1(\mathcal{Q})|}, \tag{6}$$

where $\Omega_c = \{T \subseteq X; \mu_0(T) + \mu_1(T) \geq c\}$.

We see that λ_c is small if the two distributions are similar, and the following is an example.

Example 3. *If $X = \mathbb{R}$ and $\mu_i = N(\theta_i, \sigma^2)$, then $\lambda_c \leq \frac{|\theta_0 - \theta_1|}{2\sigma c}$.*

Proof. By the total variation distance bound between two Gaussian [26, Theorem 1.3], we have $\max_{\mathcal{Q}} |\mu_0(\mathcal{Q}) - \mu_1(\mathcal{Q})| \leq \frac{|\theta_0 - \theta_1|}{2\sigma}$. The rest follows by the definition of λ_c. □

Finally, our inspection suggests that D-FA^{2}L is essentially selecting data in a special region that helps the learner to reduce model unfairness efficiently. This region is defined as follows.

Definition 4. *Let $V \subseteq H \times H$ be any hypothesis space of the decoupled model. The α-controversial region of V is*

$$\mathcal{C}_\alpha(V) = \{x \in X;\ \exists (h_0, h_1) \in V,\ |h_0(x) - h_1(x)| > \alpha\}. \tag{7}$$

The α-controversial coefficient is

$$\xi_\alpha = \sup\nolimits_{\delta > 0} \frac{\mathbb{P}_{x \sim \mu}\{\mathcal{C}_\alpha(\Sigma_{\alpha,\delta})\}}{\delta}, \tag{8}$$

where $\Sigma_{\alpha,\delta} = \{(h_0, h_1) \in H \times H;\ d_\alpha(h_0, h_1) \leq \delta\}$.

We will assume ξ_α is bounded. The following is an example.

B. Main Theoretical Results

Our first result shows that, the disparity of any decoupled model with large α-margin is 'squeezed' by their α-distance.

Theorem 5. *Let $\alpha, \gamma, c > 0$. Any model (h_0, h_1) with an α-margin of γ and $\mathbb{P}\{h_1(x) > 0.5 \mid S = 1\} \geq c$ satisfies*

$$\frac{1 - \lambda_c}{1 + \lambda_c} - \frac{d_\alpha(h_0, h_1)}{c\gamma p_0} \leq \mathtt{DI}(h_0, h_1) \leq \frac{1 + \lambda_c}{1 - \lambda_c} + \frac{d_\alpha(h_0, h_1)}{c\gamma p_0}, \tag{9}$$

where $p_0 = \mathbb{P}\{S = 0\}$.

Proof. To facilitate discussion, let us assume w.l.o.g. that X is finite so all μ_i's are probability mass functions. Write

$$\mathbb{P}\{h_i(x) > 0.5 \mid S = i\} = \sum_{x \in X} I_i^x \mu_i^x, \tag{10}$$

where $I_i^x = \mathbb{1}_{h_i(x)>0.5}$ and $\mu_i^x = \mu_i(x)$.

We first bound $\sum_x I_0^x \mu_0^x - \sum_x I_1^x \mu_1^x$, which equals to

$$\left(\sum_x I_0^x \mu_0^x - \sum_x I_1^x \mu_0^x\right) + \left(\sum_x I_1^x \mu_0^x - \sum_x I_1^x \mu_1^x\right). \tag{11}$$

The 1st parenthesized term equals to

$$\begin{aligned} \sum_x (I_0^x - I_1^x)\, \mu_0^x &\leq \sum_x \mathbb{1}_{I_0^x \neq I_1^x} \cdot \mu_0^x \\ &\leq \frac{1}{p_0} \sum_x \mathbb{1}_{I_0^x \neq I_1^x} \cdot \mu(x), \end{aligned} \tag{12}$$

where $p_0 = \mathbb{P}\{S = 0\}$. The first inequality is by case-studying I_0^x and I_1^x so that $I_0^x - I_1^x \leq \mathbb{1}_{I_0^x \neq I_1^x}$ for any x, and the second inequality is by definition so that $\mu_0(x) = \mathbb{P}\{X = x, S = 0\}/p_0 \leq \mu(x)/p_0$. Note the rightmost sum is $\mathbb{P}\{I_0^x \neq I_1^x\}$ and

$$\begin{aligned} &\mathbb{P}\{I_0^x \neq I_1^x\} \\ &= \mathbb{P}\{I_0^x \neq I_1^x,\ |h_0(x) - h_1(x)| > \alpha\} \\ &\qquad + \mathbb{P}\{I_0^x \neq I_1^x,\ |h_0(x) - h_1(x)| \leq \alpha\} \\ &\leq (1/\gamma) \cdot \mathbb{P}\{I_0^x \neq I_1^x,\ |h_0(x) - h_1(x)| > \alpha\} \\ &\leq (1/\gamma) \cdot d_\alpha(h_0, h_1), \end{aligned} \tag{13}$$

where the first inequality is due to the α-margin assumption.

For the 2nd parenthesized term in (11), since $\sum_x I_1^x \mu_1^x \geq c$,

$$\sum_x I_1^x \mu_0^x - \sum_x I_1^x \mu_1^x \leq \left(\frac{1 + \lambda_c}{1 - \lambda_c} - 1\right) \sum_x I_1^x \mu_1^x. \tag{14}$$

Putting all back to (11), we have

$$\sum_x I_0^x \mu_0^x - \left(\frac{1 + \lambda_c}{1 - \lambda_c}\right) \sum_x I_1^x \mu_1^x \ \leq\ \frac{d_\alpha(h_0, h_1)}{\gamma p_0}. \tag{15}$$

By symmetric arguments, bounding $\sum_x I_1^x \mu_1^x - \sum_x I_0^x \mu_0^x$ gives

$$\sum_x I_0^x \mu_0^x - \left(\frac{1 - \lambda_c}{1 + \lambda_c}\right) \sum_x I_1^x \mu_1^x \ \geq\ -\frac{d_\alpha(h_0, h_1)}{\gamma p_0}. \tag{16}$$

Combining (15), (16) and $\sum_x I_1^x \mu_1^x \geq c$ proves the theorem. □

The main implication of Theorem 5 is one can reduce the unfairness of a decoupled model by reducing its α-distance.

Our next theoretical result shows D-FA^{2}L can efficiently reduce the α-distance under four assumptions: **(A1)** H has a Rademacher complexity of $\mathcal{R}_m(H) \in O(1/\sqrt{m})$, **(A2)** instances are selected i.i.d. in $\mathcal{C}_\alpha(V)$, **(A3)** $d_{\alpha/2}(h_0, h_1; J) \leq 1/4\xi_\alpha$ and **(A4)** ξ_α is bounded. Assumptions (A1, A2) are common e.g., [27]. (A3) is supported by our empirical observation that $\hat{d}_{\frac{\alpha}{2}}(h_0, h_1)$ often drops to a small value.

Our second theoretical results is stated as follows.

Theorem 6. *Under (A1, A2, A3, A4), with probably at least* $1-t$, *D-FA*^{2}*L returns a model with* $d_\alpha(h_0, h_1) \leq \epsilon$ *after labeling* $O\left(\frac{\xi_\alpha^2 \cdot \log_{1/2}\epsilon}{\alpha^2} \cdot \left(32 + \sqrt{2\log\frac{4\log_{1/2}\epsilon}{t}}\right)^2\right)$ *instances.*

Proof. Let $V_j \subseteq H \times H$ be the hypothesis space satisfying (A3) on the training set before the j_{th} round of labeling. Note that$V_0 \supseteq V_1 \supseteq V_2 \ldots$. Based on (A1, A2), with probability at least $1-t'$, any decoupled model $(h_0, h_1) \in V_{j+1}$ satisfies

$$d_\alpha(h_0, h_1) \leq \hat{d}_{\frac{\alpha}{2}}(h_0, h_1; J) + \frac{K\left(32 + \sqrt{2\log(4/t')}\right)}{\alpha\sqrt{|L|}}. \quad (17)$$

The proof of (17) is similar to [28, Theorem 4.1], relying on standard Rademacher arguments plus a Lipschitz approximation of the 0-1 loss function. We defer its proof to Section VII.

Combining (17) and (A4), we see if D-FA^{2}L labels

$$N = \frac{K\xi_\alpha^2}{\alpha^2}\left(32 + \sqrt{2\log\frac{4}{t'}}\right)^2 \quad (18)$$

instances in $C_\alpha(V_j)$ in the $(j+1)_{th}$ round of labeling[1], then with probability at least $1-t'$,

$$\mathbb{P}\{|h_0(x) - h_1(x)| > \alpha \mid x \in C_\alpha(V_j)\} \leq 1/(2\xi_\alpha), \quad (19)$$

for any $(h_0, h_1) \in V_{j+1}$. This further implies

$$\begin{aligned}
&\mathbb{P}\{|h_0(x) - h_1(x)| > \alpha\} \\
&= \mathbb{P}\{|h_0(x) - h_1(x)| > \alpha \mid x \in C_\alpha(V_j)\} \cdot \mathbb{P}\{x \in C_\alpha(V_j)\} \\
&\quad + \mathbb{P}\{|h_0(x) - h_1(x)| > \alpha \mid x \notin C_\alpha(V_j)\} \cdot \mathbb{P}\{x \notin C_\alpha(V_j)\} \\
&= \mathbb{P}\{|h_0(x) - h_1(x)| > \alpha \mid x \in C_\alpha(V_j)\} \cdot \mathbb{P}\{x \in C_\alpha(V_j)\} \\
&\leq \frac{\mathbb{P}\{x \in C_\alpha(V_j)\}}{2\xi_\alpha} \\
&:= \ r_\alpha,
\end{aligned} \quad (20)$$

[1]This is obtained by setting last term in (17) to $1/4\xi_\alpha$ and solving for $|L|$.

where the second equality is by the definition of $\mathcal{C}_\alpha(V)$ so that $\mathbb{P}\{|h_0(x)-h_1(x)| > \alpha \mid x \notin C_\alpha(V_j)\} = 0$. This result implies any $(h_0, h_1) \in V_{j+1}$ falls in Σ_{α,r_α}. Thus $V_{j+1} \subseteq \Sigma_{\alpha,r_\alpha}$ and

$$\begin{aligned}
\frac{\Pr\{x \in C_\alpha(V_{j+1})\}}{\xi_\alpha} &\leq \frac{\Pr\{x \in C_\alpha(\Sigma_{\alpha,r_\alpha})\}}{\xi_\alpha} \\
&\leq r_\alpha \\
&= \frac{\Pr\{x \in C_\alpha(V_j)\}}{2\xi_\alpha},
\end{aligned} \quad (21)$$

where the second inequality is by the definition of ξ_α.

Result (21) implies that $\Pr\{C_\alpha(V_{j+1})\} \leq \frac{1}{2}\Pr\{C_\alpha(V_j)\}$ with probability at least $1-t'$. Then, after M rounds of labeling, $\mathbb{P}\{C_\alpha(V_M)\} \leq (1/2)^M$ with probability at least $1 - Mt'$. Setting $(1/2)^M = \epsilon$ gives $M = \log_{1/2}\epsilon$, and setting $Mt' = t$ gives $t' = t/M = t/\log_{1/2}\epsilon$. Putting these back to (18), we have that $d_\alpha(h_0, h_1) \leq \epsilon$ with probability at least $1-t$ after labeling $M \cdot N$ instances. This proves the theorem. ☐

Theorem 6 implies D-FA^{2}L can reduce the α-distance and the reduction is efficient under proper conditions. Combining this with Theorem 5 explains how D-FA^{2}L achieves fairness.

C. Impact of D-FA^{2}L on Model Accuracy

In the previous section, we show data training data labeled by D-FA^{2}L helps to improve model fairness. In this section, we discuss their impact on model accuracy.

Define the disagreement region w.r.t. model (h_0, h_1) as

$$\mathcal{D}(h_0, h_1) = \{x \in X; \mathbb{1}_{h_0(x)>0.5} \neq \mathbb{1}_{h_1(x)>0.5}\}, \quad (22)$$

and the controversial region w.r.t. model (h_0, h_1) as

$$\mathcal{C}_\alpha(h_0, h_1) = \{x \in X; |h_0(x) - h_1(x)| > \alpha\}. \quad (23)$$

In D-FA^{2}L, data are selected by model (h_0, h_1) uniformly at random from $\mathcal{C}_\alpha(h_0, h_1)$. On the other hand, the active learning literature says training data help to improve accuracy if selected from the following disagreement region [24, Section 2]

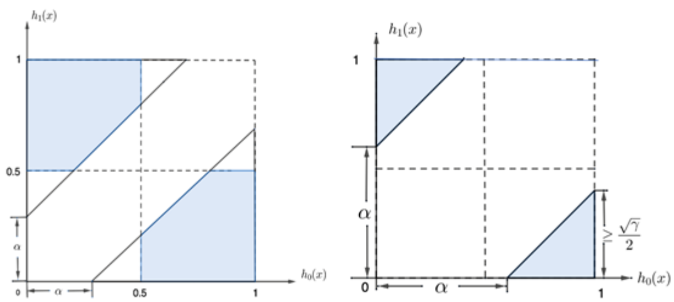
$$DIS(H) = \cup_{(h_0,h_1)\in V}\ \mathcal{D}(h_0, h_1), \quad (24)$$

where $V = H \times H$. This motivates us to come up with an insight that if $\mathcal{C}_\alpha(h_0, h_1)$ overlaps with $DIS(H)$ to a proper degree, then data selected by D-FA^{2}L could help improve accuracy with proper probability. In the following, we examine this insight through a toy example.

Figure 1a shows a population represented by their predictions received from a fixed decoupled model (h_0, h_1). Each point $(a, b) \in [0,1] \times [0,1]$ represents an instance (or, the set of instances) receiving $h_0(x) = a$ and $h_1(x) = b$. Through simple geometric arguments, we can show $\mathcal{C}_\alpha(h_0, h_1)$ is union of the upper-left and lower-right isosceles right triangles with leg length $1-\alpha$, and $\mathcal{D}(h_0, h_1)$ and is union of the upper-left and lower-right squares with side length 0.5. Their overlapping region is union of the two shaded regions.

If $\alpha < 0.5$, simple geometric arguments shows the relative area of the overlapping region is

$$\frac{|\mathcal{C}_\alpha(h_0, h_1) \cap \mathcal{D}(h_0, h_1)|}{|\mathcal{C}_\alpha(h_0, h_1)|} = \frac{0.5 - \alpha^2}{(1-\alpha)^2}, \quad (25)$$



(a) $\alpha < 0.5$ (b) $\alpha > 0.5$

Fig. 1: Toy Example

where $|\cdot|$ denotes the area of a region. Apparently, this area is larger if α is larger, implying a larger overlapping between $\mathcal{C}_\alpha(h_0, h_1)$ and $DIS(V)$. This further implies D-FA^{2}L could improve accuracy more efficiently if α is large.

If $\alpha \geq 0.5$, the overlapping region is show in Figure 1b. In this case, the shaded region always has

$$\frac{|\mathcal{C}_\alpha(h_0, h_1) \cap \mathcal{D}(h_0, h_1)|}{|\mathcal{C}_\alpha(h_0, h_1)|} = 1, \tag{26}$$

which implies

$$\mathcal{C}_\alpha(h_0, h_1) \subseteq \mathcal{D}(h_0, h_1) \subseteq DIS(H). \tag{27}$$

This implies D-FA^{2}L could improve accuracy efficiently.

V. Experiment

A. Data Preparation

We experiment on two public data sets. The COMPAS data set contains 6172 instances described by 12 numerical attributes. We treat attribute 'Two-Year-Recidivism' as label, and treat 'African-American' as the protected attribute.

The Crime and Community data set contains 1994 instances described by 128 attributes. We treat attribute 'crime rate' as the label, and treat 'percentage of African American' combined with 'percentage of African American police' as the protected attribute. We binarize the label so that a community with crime rate larger than 0.37 is considered as a high crime community, and is otherwise is considered as a low crime community. We binarize the protected attribute so that a community with less than 50% percentage sum of African American (AA) and AA police is considered a minority AA community, and is otherwise considered as a non-minority AA community.

On each data set, we randomly select 0.2% of the data as the initial labeled training set, randomly select 25% of the remaining data as the testing set, and treat the rest data as the pool of unlabeled data. We then report the average performance of each experimented method over 20 random trials. To increase numerical stability of decoupled supervised learning, in each trial, we standardize features of each group based on its selected training and pool data.

B. Experiment Design

We evaluate the performance of D-FA^{2}L and compare it with other methods. Since existing AL techniques are mostly designed for single model, we do not find any directly comparable active labeling strategy for decoupled model. Nonetheless, we adapt both classic and state-of-the-art methods and come up with the following three baseline strategies.

- Random: Data are selected uniformly at random.
- Uncertainty: It is an adaption of the classic uncertainty-based AL strategy for decoupled model. Data with the highest prediction uncertainty are selected, where the uncertainty of instance x with protected attribute $S = i$ is measured by

$$uncertainty(x) = \frac{1}{|h_i(x) - 0.5|}. \tag{28}$$

- FAL: It is an adaption of the state-of-the-art fairness-aware active labeling strategy [12] for decoupled model. First, the top m data with the highest prediction entropy are picked, where the entropy of an instance x with $S = i$ is evaluated based on $h_i(x)$. Then, among these data, the ones that could maximally reduce model unfairness are selected, where the reduction of instance with $S = i$ is measured based on h_i.

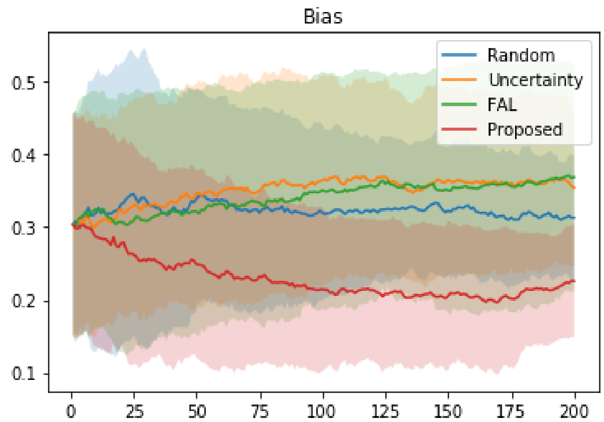
Logistic regression is used as the base model for all methods, and its regularization coefficient is set to 0.1 as it gives the best performance on initial training set. For D-FA^{2}L, we set $\alpha = 0.2$ on the COMPAS data set and $\alpha = 0.7$ on the Community and Crime data set. α can be selected as something slightly lower than the initial $|h_0(x) - h_1(x)|$. When no instance satisfies $|h_0(x) - h_1(x)| > \alpha$, we randomly pick an instance to label. For FAL, we set $m = 64$ as it is reported to give the best performance in the original paper. In each round, all methods label one instance.

We evaluate model accuracy by F1 score, and evaluate model unfairness by $\text{Bias}(h_0, h_1) = |\Pr\{h_0(x) > 0.5 \mid S = 0\} - \Pr\{h_1(x) > 0.5 \mid S = 1\}|$. Results on the two data sets are reported in Figures 2 and 3 respectively.

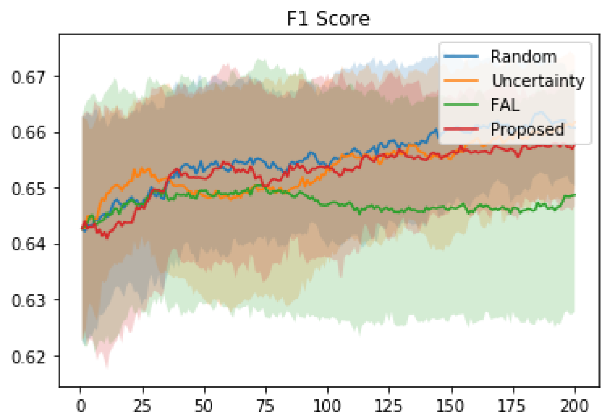
C. Discussion on the Results

Results on the COMPAS data set are reported in Figure 2. We see D-FA^{2}L is the only method that reduces model bias. In the meantime, D-FA^{2}L improves accuracy as efficiently as other strategies, suggesting it has a small trade-off between fairness and accuracy. These observations are consistent with our theoretical analysis. Results on the Community and Crime data set are shown in Figure 3. We see D-FA^{2}L effectively reduce model bias while maintaining a F1 score that is somewhere between that of the random strategy and uncertainty strategy. This further verifies the efficacy of the proposed strategy. FAL does not appear very efficient, which may imply that in the setting of decoupled model, the uncertain instances are rarely useful for improving model fairness.

When applying D-FA^{2}L, we notice that model bias keeps decreasing on the COMPAS dataset but stops decreasing after

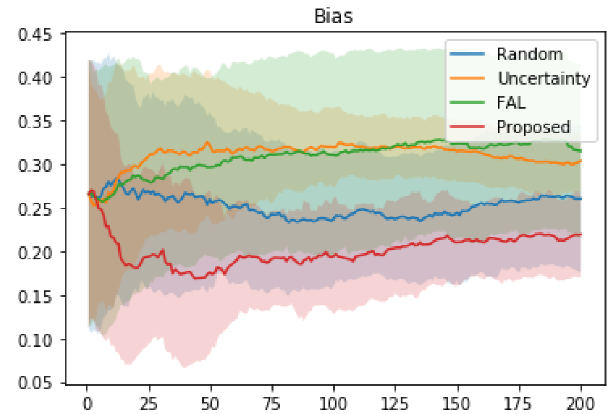


(a) Model Bias vs. Number of Samples

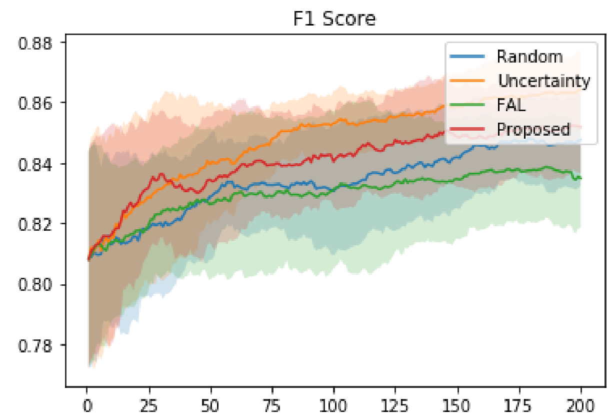


(b) F1 Score vs. Number of Samples

Fig. 2: Performance on the COMPAS Data Set



(a) Model Bias vs. Number of Samples



(b) F1 Score vs. Number of Samples

Fig. 3: Performance on the Community and Crime Data Set

about 50 rounds of labeling on the Crime dataset. This is partly related to the different distributions of $|h_0(x) - h_1(x)|$ on the two data sets. In Figure 4, we show the averaged $\Delta(x) = |h_0(x) - h_1(x)|$ of the instances labeled by D-FA^{2}L at each round. On Crime, we see $\Delta(x)$ approaches the threshold $\alpha = 0.7$ at about 50 rounds after which D-FA^{2}L loses our developed theoretical guarantee on bias reduction. On COMPAS, we see $\Delta(x)$ mostly stays above its threshold $\alpha = 0.2$ and D-FA^{2}L maintains its guarantee on bias reduction.

In Figure 5, we show the performance of D-FA^{2}L versus α. We see the bias reduction performance is not monotonic w.r.t. α, and among the examined values, $\alpha = 0.6$ achieves the highest efficiency. This makes sense: if α is too small, most instances would fall in the controversial region so selecting one from them is similar to selecting one randomly. On the other hand, if α is too large, few instances would fall in the controversial region – in this case, although D-FA^{2}L can quickly reduce bias, it will also quickly run out of data to select and start performing random selection. This explains why the red curve converges more quickly but only to a limited value in Figure 5 (a). To our surprise, α seems to have limited impact on model accuracy, allowing us to achieve a more efficient fairness-accuracy trade-off.

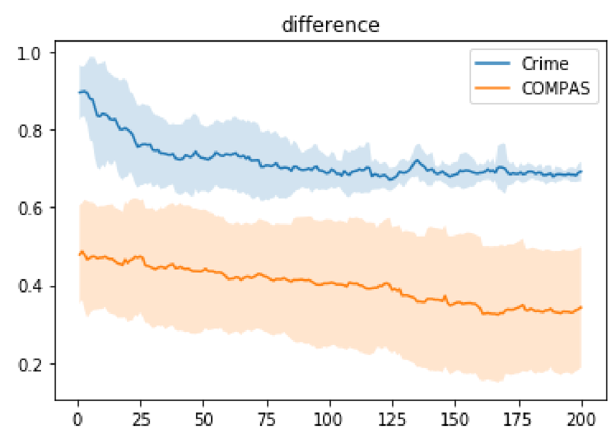
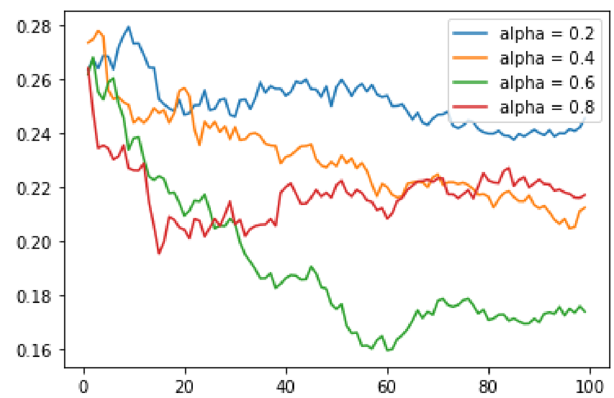


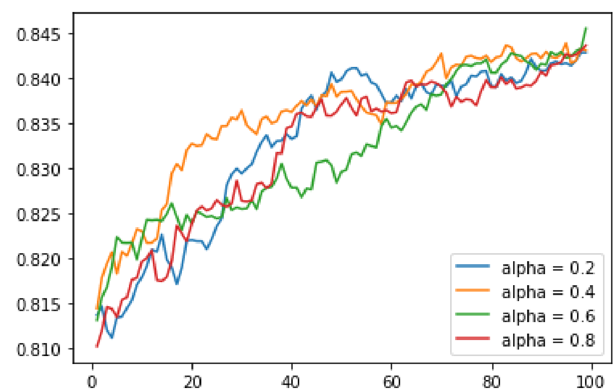
Fig. 4: $|h_0(x) - h_1(x)|$ of Instance Labeled by D-FA^{2}L

VI. Conclusion

In this paper, we study a fairly new research problem called fairness-aware active learning. We propose the first disagreement-based fairness-aware active labeling algorithm (D-FA^{2}L) for decoupled model. We first theoretically analyze D-FA^{2}L, explaining how it can reduce model unfairness effectively



(a) Model Bias decreases in different speed with different α values. To achieve best performance, α should be neither too small or too large.



(b) F1 Score increases in different speed with different α values. α seems to have limited impact on accuracy, which allows a more efficient fairness-accuracy trade-off.

Fig. 5: Performance on the Crime Data Set

and efficiently and how it impacts accuracy. We then empirically verify the efficacy of D-FA^{2}L on two data sets, showing it reduces bias more efficiently than the adapted classic and state-of-the-art methods, while maintaining comparable accuracy.

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VII. Proof

In this section, we prove the following bound in (17).

$$d_\alpha(h_0,h_1) \le \hat{d}_{\frac{\alpha}{2}}(h_0,h_1;J) + \frac{K\left(32+\sqrt{2\log(4/t')}\right)}{\alpha\sqrt{|L|}}. \quad (29)$$

We will elaborate the proof in six steps. For a function class F, let $\hat{R}_m(F)$ be its empirical Rademacher complexity over a sample J of size m, and $R_m(F) = \mathbb{E}_J[\hat{R}_m(F)]$ be its Rademacher complexity. Let σ_i be a Rademarcher noise.

Step 1. Define a set of functions

$$H' : [X] \to \{h_0(x) - h_1(x)\}_{(h_0,h_1)\in H\times H}. \quad (30)$$

We first prove $R_m(H') \le 2R_m(H)$. This is true because

$$\begin{aligned}
R_m(H') &= \mathbb{E}_J[\hat{R}_m(H')] \\
&= \mathbb{E}_{J,\sigma}\left[\sup_{(h_0,h_1)\in H^2}\left(\frac{1}{m}\sum_{i=1}^m \sigma_i(h_0(x_i)-h_1(x_i))\right)\right] \\
&\le \mathbb{E}_{J,\sigma}\left[\sup_{h_0\in H}\left(\frac{1}{m}\sum_{i=1}^m \sigma_i h_0(x_i)\right)\right. \\
&\quad \left. + \sup_{h_1\in H}\left(\frac{1}{m}\sum_{i=1}^m -\sigma_i h_1(x_i)\right)\right] \\
&= \mathbb{E}_{J,\sigma}\left[\sup_{h_0\in H}\left(\frac{1}{m}\sum_{i=1}^m \sigma_i h_0(x_i)\right)\right] \\
&\quad + \mathbb{E}_{J,\sigma}\left[\sup_{h_1\in H}\left(\frac{1}{m}\sum_{i=1}^m -\sigma_i h_1(x_i)\right)\right] \\
&= 2R_m(H).
\end{aligned} \quad (31)$$

Step 2. Let $abs()$ be the absolute value function. Define a set of composed functions:

$$G := abs \circ H' : [X] \to \{|h_0(x) - h_1(x)|\}_{(h_0,h_1)\in H\times H}. \quad (32)$$

We can prove $R_m(G) \le 4R_m(H)$. This is true because

$$R_m(G) = R_m(abs\circ H') \le 2R_m(H') \le 4R_m(H), \quad (33)$$

where the first inequality is based on [27, Theorem 12, Fact 4] and the fact that absolute value function has a Lipschitz constant 1, and the second inequality is based on (31).

Step 3. Define a composed function

$$\tilde{F}_\alpha := \tau_\alpha^t \circ G, \quad (34)$$

where

$$\tau_\alpha^t(x) = \begin{cases} 0, & x \le \alpha \\ t(x-\alpha), & \alpha < x < \alpha + \frac{1}{t} \\ 1, & x \ge \alpha + \frac{1}{t}, \end{cases} \quad (35)$$

We can prove $R_m(\tilde{F}) \le 8\Delta R_m(H)$. This is true because

$$R_m(\tilde{F}) \le 2tR_m(G) \le 8tR_m(H), \quad (36)$$

where the first inequality is based on [27, Theorem 12, Fact 4] and the fact that τ_α^t has a Lipschitz constant t; the second inequality is based on (33).

Step 4. Let $\tau_\alpha(x)$ be an indicator function outputting 1 if $x > \alpha$ and 0 otherwise. There is

$$\tau_{\alpha+\frac{1}{t}}(z) \le \tau_\alpha^t(z) \le \tau_\alpha(z), \quad (37)$$

for any input z. Define loss functions for a given (h_0,h_1) as

$$\mathcal{L}_\alpha^t(h_0,h_1) = \mathbb{E}_x \tau_\alpha^t(|h_0(x)-h_1(x)|), \quad (38)$$

and, similarly,

$$\mathcal{L}_\alpha(h_0,h_1) = \mathbb{E}_x \tau_\alpha(|h_0(x)-h_1(x)|). \quad (39)$$

Let $\hat{\mathcal{L}}_\alpha^t$ and $\hat{\mathcal{L}}_\alpha$ be the empirical estimates of $\mathcal{L}_\alpha^t$ and $\mathcal{L}_\alpha$ over sample S, respectively.

Combining (37, 38,39), we can easily show that

$$\mathcal{L}_{\alpha+\frac{1}{t}}(h_0,h_1) \le \mathcal{L}_\alpha^t(h_0,h_1) \le \mathcal{L}_\alpha(h_0,h_1), \quad (40)$$

and

$$\hat{\mathcal{L}}_{\alpha+\frac{1}{t}}(h_0,h_1) \le \hat{\mathcal{L}}_\alpha^t(h_0,h_1) \le \hat{\mathcal{L}}_\alpha(h_0,h_1). \quad (41)$$

Further, by a classic error bound e.g., [29, Theorem 3.1], for any (h_0,h_1), with probability at least $1-\delta$, there is

$$\begin{aligned}
\left|\mathcal{L}_\alpha^t(h_0,h_1) - \hat{\mathcal{L}}_\alpha^t(h_0,h_1)\right| &\le 2R_m(\tilde{F}) + \sqrt{\frac{\ln(4/\delta)}{2m}} \\
&\le 8tR_m(H) + \sqrt{\frac{\ln(4/\delta)}{2m}}.
\end{aligned} \quad (42)$$

where the second inequality is based on (36).

Step 5. Assuming $t \ge 1$, it follows

$$\begin{aligned}
&\hat{\mathcal{L}}_\alpha(h_0,h_1) \\
&\ge \hat{\mathcal{L}}_\alpha^t(h_0,h_1) \\
&\ge \mathcal{L}_\alpha^t(h_0,h_1) - \left(8tR_m(H) + \sqrt{\frac{\ln(4/\delta)}{2m}}\right) \\
&\ge \mathcal{L}_\alpha^t(h_0,h_1) - t\left(8R_m(\mathcal{H}) + \sqrt{\frac{\ln(4/\delta)}{2m}}\right) \\
&\ge \mathcal{L}_{\alpha+\frac{1}{t}}(h_0,h_1) - t\left(8R_m(\mathcal{H}) + \sqrt{\frac{\ln(4/\delta)}{2m}}\right),
\end{aligned} \quad (43)$$

where the first and last inequalities are by (41), the second is based on (42), and the third is based on $t \ge 1$.

Step 6. By the assumption that $R_m(H) \in O(1/\sqrt{m})$, there exists a constant $C > 0$ such that, for any $(h_0,h_1) \in H \times H$,

$$\begin{aligned}
&\mathbb{P}\{|h_0(x)-h_1(x)| > \alpha + 1/t\} \\
&= \mathcal{L}_{\alpha+\frac{1}{t}}(h_0,h_1) \\
&\le \hat{\mathcal{L}}_\alpha(h_0,h_1) + t\left(8R_m(H) + \sqrt{\frac{\ln(4/\delta)}{2m}}\right) \\
&\le \hat{\mathcal{L}}_\alpha(h_0,h_1) + t\left(\frac{8C}{\sqrt{m}} + \sqrt{\frac{\ln(4/\delta)}{2m}}\right),
\end{aligned} \quad (44)$$

where the first inequality is based on (43).

Recall $\alpha \in (0,1)$. Setting $\frac{1}{t} = \alpha$, we have

$$\begin{aligned} &\mathbb{P}\{|h_0(x) - h_1(x)| > 2\alpha\} \\ &\qquad \leq \hat{\mathcal{L}}_\alpha(h_0, h_1) + \frac{1}{\alpha}\left(\frac{8C}{\sqrt{m}} + \sqrt{\frac{\ln(4/\delta)}{2m}}\right). \end{aligned} \tag{45}$$

Setting $\alpha' = 2\alpha$ proves (17).